

Twitter Clone (Full Stack)

Project Objective

Build a **Twitter-like social media web application** that allows users to create profiles, post tweets, and interact with other users using modern full-stack technologies.



Tech Stack (Mandatory)

- **Frontend:** EJS (Embedded JavaScript Templates)
 - **Backend:** Node.js with Express.js
 - **Database:** MySQL
 - **Authentication:** JWT (**Mandatory**)
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Core Features

1. Authentication & Security (Very Important)

You must implement a complete authentication system.

Required:

- User Registration (Signup)
- Login
- Logout
- **JWT-based Authentication (Mandatory)**
- Unique Username validation
- Password hashing
- Protected routes/pages for authenticated users

Additional Required Features:

- Forgot Password
- OTP-based Password Reset
- Remember Me functionality

- Change Password
 - Email Verification (recommended)
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Forgot Password with OTP (Mandatory)

Trainees must implement OTP-based password reset.

Important Change:

- **Do NOT use SMTP or Nodemailer**
 - Instead, create a **custom page that simulates an email template**
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Email Simulation Requirement

Instead of sending real emails:

- Create a **UI page that looks like an email inbox/message**
- Display the OTP inside a **designed email template UI**
- This page should simulate how a real email looks

Example:

After user enters email:

- Redirect to a page like:
 - 👉 “Your Email Inbox (Simulation)”
 - Show:
 - Subject: *Password Reset Request*
 - Message body with OTP code
 - Styled like a real email
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Required OTP Rules

- OTP should remain valid for **2 minutes only**
- A **countdown timer** must be shown
- **Resend OTP button should appear only after 2 minutes**
- Expired OTP must not be accepted

- New OTP should invalidate the previous OTP
 - Show proper success and error messages
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Flow

1. User clicks **Forgot Password**
 2. User enters registered email
 3. System generates OTP
 4. Redirect to **Email Simulation Page (EJS view)**
 5. Show OTP inside email-like UI
 6. Start **2-minute timer**
 7. User enters OTP
 8. If valid → allow password reset
 9. If expired → enable **Resend OTP**
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2. User Profile Management

Each user should have a profile with:

- Profile Picture
- Cover Image
- Full Name
- Unique Username
- Bio
- Email
- Joined Date
- **Time Zone (Mandatory)**

Features:

- Create Profile
 - Edit Profile
 - Update Profile Picture
 - Update Cover Image
 - View Own Profile
 - View Other Users' Profiles
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3. Tweet System

Users should be able to:

- Create Tweet
- View Tweets
- Delete Own Tweet
- Edit Tweet (optional)
- Add text content
- Add image (optional but recommended)

Each tweet may contain:

- Text content
 - Image
 - Author details
 - Timestamp
 - Like count
 - Comment count
 - Repost count
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4. Social Interaction

Implement core Twitter-style interactions:

- Like Tweet
 - Comment on Tweet
 - Repost / Retweet
 - Share Tweet
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5. Follow System

Users can:

- Follow other users
 - Unfollow other users
 - View followers
 - View following
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6. Feed / Timeline

The application should display a timeline/feed that includes:

- User's own tweets
 - Tweets from followed users
 - Latest tweets in proper order
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7. Search Functionality

Users should be able to search:

- Users by name or username
 - Tweets by text/keyword (optional)
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8. Media Upload

Support image upload for:

- Profile Picture
- Cover Image
- Tweet Image

You may use:

- Local storage for basic version
 - Cloud storage service for advanced version
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Functional Requirements

The application should allow users to:

- Register and login securely
- Use JWT authentication
- Use Remember Me
- Reset password using OTP
- Verify OTP within 2 minutes
- Resend OTP only after timer expiry

- Create and manage profile
 - Post tweets
 - Like, comment, repost, and share tweets
 - Follow/unfollow other users
 - View feed and profiles
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Non-Functional Requirements

- Clean and responsive UI
- Secure authentication flow
- Proper validation
- Proper error handling
- Modular and clean code structure
- Reusable components
- RESTful backend structure
- Good coding practices